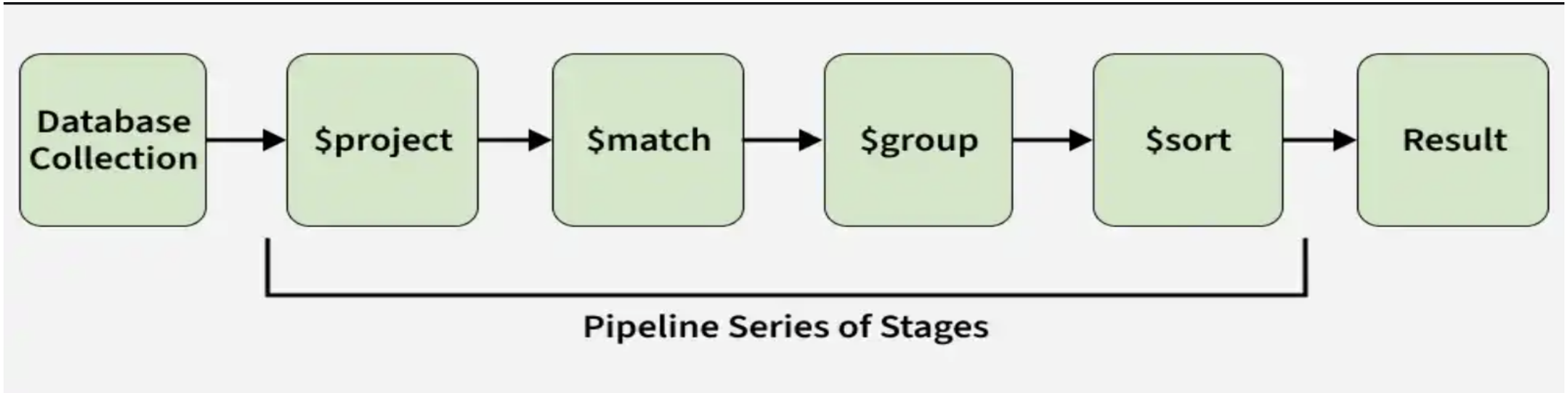
# 24CSA212 – Full Stack Frameworks

## AGGREGATION FRAMEWORK

# Aggregation in MongoDB

- MongoDB Aggregation processes documents through a pipeline of stages to transform, analyze, and compute results directly in the database.
- Processes data step by step through pipeline stages.
- Performs computations on the server to reduce client load.
- Transforms and reshapes documents for analytics.



# Aggregation Approaches

- MongoDB provides multiple approaches for performing aggregation -pending on the complexity and type of data analysis you need to perform.

- **1. Single Purpose Aggregation**
- **2. MongoDB Aggregation Pipeline**

# Single Purpose Aggregation

- Single-purpose aggregation methods are used for quick, simple analytics like counting documents or fetching distinct field values, without using the aggregation pipeline.

- **count():** Returns the number of documents in a collection.
- **distinct():** Retrieves unique values for a specified field.
- **estimatedDocumentCount():** Provides an estimated count of documents.

# Single Purpose Aggregation

- **Example:** Counting Users in Each City
- Consider a single-purpose aggregation example where we find the total number of users in each city from the users collection.

**Explanation**

- Groups documents by the city field.
- Uses $sum to count users in each city.
- Returns documents with city (_id) and totalUsers.

```
db.users.aggregate([
  { $group: { _id: "$city", totalUsers: { $sum: 1 } } }
])
```

**Output:**

```
[
  { _id: 'Los Angeles', totalUsers: 1 },
  { _id: 'New York', totalUsers: 1 },
  { _id: 'Chicago', totalUsers: 1 }
]
```

# MongoDB Aggregation Pipeline

- The MongoDB aggregation pipeline is a multi-stage process where each stage transforms documents.
- The output of one stage becomes the input for the next, with each stage filtering, modifying, or computing on documents until the final result is produced.
- Filters that will operate like queries.
- The document transformation that modifies the resultant document.
- Provide pipeline tools for grouping and sorting documents.
- Aggregation pipeline can also be used in sharded collection.

# MongoDB Aggregation Pipeline

The aggregate() function is used to perform aggregation. It can have three operators stages, expression and accumulator. These operators work together to achieve final desired outcome.

```
db.collection.aggregate([
  {
    $group: {
      _id: "$groupField",
      totalCount: { $sum: 1 }
    }
  }
])
```

- **Stages:** Define each step in the aggregation pipeline to filter, group, sort, or reshape documents.
- **Expressions:** Compute values and transform fields within pipeline stages.
- **Accumulators:** Perform calculations (sum, average, count) while grouping documents.

# Aggregation Pipeline Method

**1. $group:** It Groups documents by the city field and calculates the average age using the $avg accumulator.

.

```
db.users.aggregate([
  { $group: { _id: "$city", averageAge: { $avg: "$age" } } }
])
```

**Output:**

```
[
  { _id: 'New York', averageAge: 30 },
  { _id: 'Chicago', averageAge: 25 },
  { _id: 'Los Angeles', averageAge: 35 }
]
```


```
{
  "_id": ObjectId("60a3c7e96e06f64fb5ac0700"),
  "name": "Alex",
  "age": 30,
  "email": "alex@example.com",
  "city": "New York"
}
{
  "_id": ObjectId("60a3c7e96e06f64fb5ac0701"),
  "name": "Ben",
  "age": 35,
  "email": "ben@example.com",
  "city": "Los Angeles"
}
{
  "_id": ObjectId("60a3c7e96e06f64fb5ac0702"),
  "name": "Clevin",
  "age": 25,
  "email": "clevin@example.com",
  "city": "Chicago"
}
```

# Aggregation Pipeline Method

**2. $project:** Include or exclude fields from the output documents.

```
db.users.aggregate([
  { $project: { name: 1, city: 1, _id: 0 } }
])
```

**Output:**

```
[
  { name: 'Alex', city: 'New York' },
  { name: 'Ben', city: 'Los Angeles' },
  { name: 'Clevin', city: 'Chicago' }
]
```

```
{
  "_id": ObjectId("60a3c7e96e06f64fb5ac0700"),
  "name": "Alex",
  "age": 30,
  "email": "alex@example.com",
  "city": "New York"
}
{
  "_id": ObjectId("60a3c7e96e06f64fb5ac0701"),
  "name": "Ben",
  "age": 35,
  "email": "ben@example.com",
  "city": "Los Angeles"
}
{
  "_id": ObjectId("60a3c7e96e06f64fb5ac0702"),
  "name": "Clevin",
  "age": 25,
  "email": "clevin@example.com",
  "city": "Chicago"
}
```

# Aggregation Pipeline Method

**3. $match:** Filter documents to pass only those that match the specified condition(s).

```
db.users.aggregate([
  { $match: { age: { $gt: 30 } } }
])
```

**Output:**

```
[
  {
    _id: ObjectId('60a3c7e96e06f64fb5ac0701'),
    name: 'Ben',
    age: 35,
    email: 'ben@example.com',
    city: 'Los Angeles'
  }
]
```

```
{
  "_id": ObjectId("60a3c7e96e06f64fb5ac0700"),
  "name": "Alex",
  "age": 30,
  "email": "alex@example.com",
  "city": "New York"
}
{
  "_id": ObjectId("60a3c7e96e06f64fb5ac0701"),
  "name": "Ben",
  "age": 35,
  "email": "ben@example.com",
  "city": "Los Angeles"
}
{
  "_id": ObjectId("60a3c7e96e06f64fb5ac0702"),
  "name": "Clevin",
  "age": 25,
  "email": "clevin@example.com",
  "city": "Chicago"
}
```

# Aggregation Pipeline Method

**4. $sort:** It Sorts documents based on field values.

```
db.users.aggregate([
  { $sort: { age: 1 } }
])
```

**Output:**

```
[
  {
    _id: ObjectId('60a3c7e96e06f64fb5ac0702'),
    name: 'Clevin',
    age: 25,
    email: 'clevin@example.com',
    city: 'Chicago'
  },
  {
    _id: ObjectId('60a3c7e96e06f64fb5ac0700'),
    name: 'Alex',
    age: 30,
    email: 'alex@example.com',
    city: 'New York'
  },
  {
    _id: ObjectId('60a3c7e96e06f64fb5ac0701'),
    name: 'Ben',
    age: 35,
    email: 'ben@example.com',
    city: 'Los Angeles'
  }
]
```


```
{
  "_id": ObjectId("60a3c7e96e06f64fb5ac0700"),
  "name": "Alex",
  "age": 30,
  "email": "alex@example.com",
  "city": "New York"
}
{
  "_id": ObjectId("60a3c7e96e06f64fb5ac0701"),
  "name": "Ben",
  "age": 35,
  "email": "ben@example.com",
  "city": "Los Angeles"
}
{
  "_id": ObjectId("60a3c7e96e06f64fb5ac0702"),
  "name": "Clevin",
  "age": 25,
  "email": "clevin@example.com",
  "city": "Chicago"
}
```

# Aggregation Pipeline Method

**5. $limit:** Limit the number of documents passed to the next

```
db.users.aggregate([
  { $limit: 2 }
])
```

**Output:**

```
[
  {
    _id: ObjectId('60a3c7e96e06f64fb5ac0700'),
    name: 'Alex',
    age: 30,
    email: 'alex@example.com',
    city: 'New York'
  },
  {
    _id: ObjectId('60a3c7e96e06f64fb5ac0701'),
    name: 'Ben',
    age: 35,
    email: 'ben@example.com',
    city: 'Los Angeles'
  }
]
```

```
{
  "_id": ObjectId("60a3c7e96e06f64fb5ac0700"),
  "name": "Alex",
  "age": 30,
  "email": "alex@example.com",
  "city": "New York"
}
{
  "_id": ObjectId("60a3c7e96e06f64fb5ac0701"),
  "name": "Ben",
  "age": 35,
  "email": "ben@example.com",
  "city": "Los Angeles"
}
{
  "_id": ObjectId("60a3c7e96e06f64fb5ac0702"),
  "name": "Clevin",
  "age": 25,
  "email": "clevin@example.com",
  "city": "Chicago"
}
```

# Using MongoDB Aggregation

- To use MongoDB for aggregating data, follow below steps:
- **1. Connect to MongoDB:** Ensure you are connected to your MongoDB instance.
- **2. Choose the Collection:** Select the collection you want to perform aggregation on, such as students.
- **3. Define the Aggregation Pipeline:** Create an array of stages, like $group to group documents and perform operations (e.g., calculate the average grade).
- **4. Run the Aggregation Query:** Use the aggregate method on the collection with your defined pipeline.
- **Example:** Calculates the average grade of all students in the students collection.

```
db.students.aggregate([
  {
    $group: {
      _id: null,
      averageGrade: { $avg: "$grade" }
    }
  }
])
```

**Output:**

```
[
  { "_id": null, "averageGrade": 85 }
]
```

# Factors affecting Aggregation performance in MongoDB

- Aggregation speed depends on pipeline complexity, data size, server hardware, and index efficiency.
- MongoDB’s aggregation framework is designed to handle large data sets and complex operations efficiently.
- Proper query optimization and use of indexes improve performance.
- Server configuration plays a key role in ensuring fast and scalable aggregation.
- Performance can vary by use case and setup, so monitoring and tuning are important.

# Aggregation Approaches